

Gerasimos Kouniatalis

Theoretical Cosmology | Early-Universe Physics | Gravitation

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Research Profile

PhD researcher in theoretical cosmology working across inflation, quantum cosmology, modified gravity, black-hole physics and cosmological tensions. My research combines analytic model building with gravitational dynamics and phenomenology, with particular emphasis on connecting fundamental physics to falsifiable cosmological or strong-gravity signatures. I have **7 peer-reviewed publications (5 first-author)**, including papers in *JCAP*, *Physical Review D*, *Physics Letters B*, *Physics of the Dark Universe* and *European Physical Journal C*, together with an accepted cosmology manuscript and recent preprints on inflationary initial conditions, quantum cosmology and the Hubble tension.

Research Interests

Inflation and its initial conditions; quantum cosmology and the Wheeler–DeWitt equation; primordial black holes and early-Universe dynamics; modified gravity and geometric extensions of GR; black-hole lensing and quasinormal modes; cosmological tensions and model-independent tests of the expansion history.

Education

Ongoing **PhD Candidate in Theoretical Cosmology**, National Technical University of Athens & National Observatory of Athens. Thesis: *Investigating gravity in light of modern cosmological tensions*.
MSc in Nuclear and Particle Physics, National and Kapodistrian University of Athens, Department of Physics. Thesis: *Weyl Geometries and Holographic Renormalization in AlAdS Manifolds*. **Grade: 10/10**.
BSc in Mathematics, National and Kapodistrian University of Athens, Department of Mathematics. Track: Theoretical Mathematics.

Research Experience & Scientific Service

- **PhD Researcher, NTUA / National Observatory of Athens:** early-Universe cosmology, modified gravity, quantum cosmology, black-hole perturbations, gravitational lensing and cosmological phenomenology.
- **Ad hoc referee:** *Physics of the Dark Universe*, *Physics Letters B*, *Nuclear Physics B*, and *Journal of Modern Physics*.
- Contributor to the **CosmoVerse White Paper** on observational tensions in cosmology (*Phys. Dark Univ.* 49, 101965, 2025), including work connected with modified-gravity approaches to cosmological tensions.

Research Highlights

- Developed a ghost-free Weyl-geometric modified-gravity construction with second-order field equations and cosmological applications.
- Built inflationary models with strongly suppressed primordial tensor amplitudes while retaining viable slow-roll dynamics and reheating.
- Connected quantum-cosmological wave functions and Bohmian trajectories to cosmological expansion, and developed a dynamical mechanism for preparing plateau-inflation initial conditions.
- Investigated observable strong-gravity signatures through black-hole lensing, geodesic instability and quasinormal-mode decay, and early-Universe signatures through primordial-black-hole formation.

Technical Skills

Scientific computing/typesetting: Mathematica, MATLAB, L^AT_EX; symbolic and numerical modelling; finite-difference methods. **Languages:** Greek, English, French.

Selected Publications

- G. Kouniatalis**, “Initial conditions for inflation,” [arXiv:2609.09200](#) (2026).
- G. Kouniatalis**, T. Papanikolaou, S. Basilakos, E. N. Saridakis, “Primordial black hole formation from transient $f(T)$ cosmology,” *Phys. Dark Univ.* **53**, 102418 (2026), [doi:10.1016/j.dark.2026.102418](#).
- S. Basilakos, **G. Kouniatalis**, E. N. Saridakis, C. Tzerefos, “Bohmian quantum cosmology from the Wheeler–DeWitt equation,” *Phys. Lett. B* **875**, 140339 (2026), [doi:10.1016/j.physletb.2026.140339](#).
- G. Kouniatalis**, “A unified scalar-field resolution of the H_0 , S_8 and evolving Dark Energy tensions,” accepted in *Astronomy* (2026), [arXiv:2603.11706](#).
- G. Kouniatalis**, E. N. Saridakis, “A null test of the Hubble tension,” [arXiv:2601.22043](#) (2026).
- G. Kouniatalis**, “The Wave Function of the Universe and Inflation,” [arXiv:2510.04775](#) (2025; revised 2026).
- G. Kouniatalis**, A. G. Suvorov, K. Destounis, “Lensing by black holes within astrophysical environments,” *Phys. Rev. D* **112**, 124062 (2025), [doi:10.1103/tn1g-7tfj](#).
- G. Kouniatalis**, P. A. González, E. Papantonopoulos, Y. Vásquez, “Near-horizon geodesic instabilities and anomalous decay of quasinormal modes in Weyl black holes,” *Phys. Rev. D* **112**, 124061 (2025), [doi:10.1103/ftvr-pclv](#).
- G. Kouniatalis**, E. N. Saridakis, “Inflation from a generalized exponential plateau: towards extra suppressed tensor-to-scalar ratios,” *JCAP* **11**, 038 (2025), [doi:10.1088/1475-7516/2025/11/038](#).
- G. Kouniatalis**, E. N. Saridakis, “Modified gravity from Weyl connection and the $f(R, \mathcal{A})$ extension,” *Eur. Phys. J. C* **85**, 260 (2025), [doi:10.1140/epjc/s10052-025-14017-z](#).

Complete and updated publication list supplied separately.

Teaching & Scientific Communication

- Mathematics and physics teaching/tutoring, including special relativity, general relativity and cosmology; previous mathematics teaching appointment at Ionios School.
- **Mensa International member with the highest possible score**; delivered lectures for Mensa including “Proof of planets’ elliptic orbit” (2022) and “Cosmology: The Art of Understanding the Whole” (2025), alongside seminars for advanced/gifted students on modern physics.